The Riemann Hypothesis via the Hilbert-Pólya Conjecture

Diego Rincón

Independent

Abstract

Every nontrivial zero $\rho = \sigma + i\gamma$ of the Riemann zeta function departs from the critical line by $\delta = \sigma - 1/2$. The Riemann Hypothesis is the claim $\delta = 0$. The Hilbert-Pólya conjecture proposes a self-adjoint operator H on a Hilbert space whose spectrum equals the imaginary parts of the nontrivial zeros; self-adjointness immediately forces $\delta = 0$. We prove RH $\Leftarrow$ HP in one line and survey the evidence for HP. The remaining open work is the construction of H. In terms of the adaptive regularization template: the regularize, compact, and regularity steps are vacuous once H exists; the problem is entirely in the bound step (spectral control), which is the construction problem itself.

1 The Departure and the Critical Line

Let $\rho = \sigma + i\gamma$ be a nontrivial zero of $\zeta(s)$. The *departure from the critical line* is $\delta = \sigma - 1/2$.

Riemann Hypothesis (RH). $\delta = 0$ for all nontrivial zeros.

The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$. Zeros come in pairs $(\rho, 1-\bar{\rho})$ symmetric about $\mathrm{Re}(s) = 1/2$. On the critical line $s = 1/2 + it$: $\overline{\xi(s)} = \xi(\bar{s}) = \xi(1-s) = \xi(s)$, so ξ is real-valued and zeros on the critical line are real zeros of a real analytic function of t.

Known unconditionally: all nontrivial zeros have $0 < \sigma < 1$ (classical); a positive proportion lie on the critical line (Selberg, Hardy-Littlewood); the first 10^{13} zeros verified on the critical line by numerical computation.

2 The Hilbert-Pólya Conjecture

Conjecture 1 (Hilbert-Pólya, ca. 1914; see [4]). *There exists a self-adjoint operator H on a Hilbert space $\mathcal{H}$ such that*

$$\zeta\left(\tfrac{1}{2} + i\lambda\right) = 0 \iff \lambda \in \mathrm{Spec}(H).$$

Theorem 2 (HP implies RH). *Conjecture 1 implies the Riemann Hypothesis.*

Proof. A self-adjoint operator has real spectrum. If $\lambda \in \mathrm{Spec}(H)$ and $\zeta(1/2 + i\lambda) = 0$, then the zero is $\rho = 1/2 + i\lambda$ with $\sigma = 1/2$, so $\delta = 0$. $\square$

Remark 3. The implication HP $\Rightarrow$ RH requires no computation beyond the definition of self-adjoint. The content of RH is entirely contained in HP.

3 Adaptive Regularization Frame

For completeness, we note why the adaptive regularization template (regularize $\to$ bound $\to$ compact $\to$ regularity) collapses for RH.

Given a candidate operator $\widetilde{H}$ approximating the Hilbert-Pólya operator, define $\widetilde{H}_\alpha = (\widetilde{H} + \widetilde{H}^*)/2 + \alpha\widetilde{H}$ for $\alpha > 0$. As $\alpha \to 0^+$, $\widetilde{H}_\alpha$ interpolates from the Hermitian part toward $\widetilde{H}$. The Hermitian part $(\widetilde{H} + \widetilde{H}^*)/2$ is self-adjoint by construction and has real spectrum. But its spectrum need not match the zeros of $\zeta(s)$, since projecting to the Hermitian part distorts the eigenvalues.

The difficulty is not the regularization: any self-adjoint operator has real spectrum trivially. The difficulty is finding an operator — regularized or not — *whose spectrum is the imaginary parts of the zeros*. That is the construction problem, and it is not addressed by regularization.

Remark 4 (Template status).

- *Regularize*: trivial (any self-adjoint operator works). ✓
- *Bound*: spectral control connecting the operator to the zeros. **Open.**
- *Compact*: spectral continuity in α. ✓ (once operator exists)
- *Regularity*: $\delta = 0$ from self-adjointness. ✓ (one line)

The problem is entirely in the bound step: constructing the operator.

4 Evidence for the Hilbert-Pólya Conjecture

GUE statistics. Montgomery's pair correlation conjecture [2] predicts that spacings between consecutive zeros follow the distribution $1-(\sin \pi u/\pi u)^2$ in the limit. This is precisely the pair correlation of eigenvalues of a random matrix from the Gaussian Unitary Ensemble (GUE) — the ensemble of large Hermitian matrices with Gaussian entries. Odlyzko's numerical computations [3] confirm this to high precision. GUE is the eigenvalue distribution of a *generic self-adjoint operator*, strongly suggesting H exists and is self-adjoint.

Berry-Keating. Berry and Keating [4] propose $H = xp + px$ (quantization of the classical Hamiltonian xp) on a suitable half-line with appropriate boundary conditions. The classical orbits of $H_{\mathrm{cl}} = xp$ are hyperbolic and give the right leading-order zero density via the Gutzwiller trace formula. The boundary conditions that force quantization to the zeta zeros remain under investigation.

Connes. Connes [5] constructs an operator on an adelic L^2 space with a formal trace formula reproducing the explicit formula for ζ. The zeros appear as *absorption spectrum* (missing eigenvalues) rather than as eigenvalues; making this precise as a Hilbert-Pólya operator in the classical sense is an open problem.

Selberg zeta functions. For compact hyperbolic surfaces, the Selberg zeta function has zeros at $s = 1/2 + i\lambda_n$ where $\lambda_n^2 + 1/4$ are eigenvalues of the Laplacian Δ — a genuine self-adjoint operator. This is the prototype for HP: the Selberg case proves the analogue of RH automatically. The Riemann case is harder because no hyperbolic surface with Laplacian spectrum matching the Riemann zeros is known.

5 What Remains

The Riemann Hypothesis is equivalent, via Theorem 2, to the Hilbert-Pólya conjecture. Proving HP requires constructing a self-adjoint operator H on an explicit Hilbert space with $\mathrm{Spec}(H) = \{\gamma : \zeta(1/2 + i\gamma) = 0\}$. The GUE statistics, Berry-Keating, and Connes programs each provide partial evidence or heuristic constructions. None has produced a proven self-adjoint operator matching the zeros.

The standard alternative approaches — zero-free regions, explicit formulas, L-function analogues, zero-density estimates — are orthogonal to the Hilbert-Pólya strategy and equally incomplete.

References

[1] Riemann, B. (1859). Über die Anzahl der Primzahlen unter einer gegebenen Größe. *Monatsberichte der Berliner Akademie.*

[2] Montgomery, H. (1973). The pair correlation of zeros of the zeta function. *Proc. Symp. Pure Math.* 24, 181–193.

[3] Odlyzko, A. (1987). On the distribution of spacings between zeros of the zeta function. *Math. Comp.* 48, 273–308.

[4] Berry, M., Keating, J. (1999). The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41, 236–266.

[5] Connes, A. (1999). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.* 5, 29–106.

[6] Kato, T. (1966). *Perturbation Theory for Linear Operators.* Springer.